

USE CASE “CUD PRODUCT”

1. Use case code

UC004

2. Brief description

This use case allows the Product manager to maintain the product catalog. It includes adding new physical media (Books, CDs, DVDs, Newspapers), updating existing product information, and removing products from the system according to stock and quantity constraints.

3. Actors: Product Manager

4. Preconditions

- Product manager is logged into the AIMS software.
- The software is displaying the product management dashboard.

5. Flow of events

5.1. Basic flow

5.1.1. Basic flow: Create product

Step 1: Product manager selects the "Add product" option.

Step 2: AIMS software displays the product entry form (Title, Category, Price, Weight, etc.).

Step 3: Product manager fills in the form and selects the product type (Book, CD, DVD, or Newspaper). (see Table 2)

Step 4: AIMS software adds specific fields for that type. (see Table 3, 4, 5, 6)

Step 5: Product manager enters all required information and clicks "Save".

Step 6: AIMS software validates the data and uniqueness of the barcode.

Step 7: AIMS software saves the new product and displays a success message.

5.1.2. Basic flow: Update product

Step 1: Product manager selects a product from the list and clicks "Edit".

Step 2: AIMS software retrieves and displays the current information of the product.
(see Table 7 and Table 9, 10, 11, 12, 13)

Step 3: Product manager modifies the desired fields.

Step 4: Product manager clicks "Update".

Step 5: AIMS software validates the changes.

Step 6: AIMS software saves updated information and notifies the Product manager.

5.1.3. Basic flow: Delete product

Step 1: Product manager selects one or more products (up to 10) from the list.
(see Table 8)

Step 2: Product manager clicks "Delete".

Step 3: AIMS software checks the stock level for each selected product.

Step 4: For products with stock = 0, AIMS software permanently deletes them.

Step 5: For products with stock > 0, AIMS software changes their status to "Deactivated".

Step 6: AIMS software displays a summary of the action (number of products deleted vs. deactivated).

5.2. Alternative flow

No	Location	Condition	Action	Resume location
1	At step 6 (Create) or step 5 (Update)	AIMS software detects invalid formats (e.g., negative price, empty title).	Action 1: AIMS software highlights the error fields. Action 2: The Product manager corrects the data and re-submits.	At step 7 (Create) or step 6 (Update)
2	At step 1 (Delete)	Product manager selects more than 10 products.	AIMS software prevents the action and displays failure message.	At step 1(Delete)
3	At step 2 (Delete)	Total deleted products of the day have already been 20.	AIMS software prompts a failure message of exceeding daily quota (20)	Use case ends

4	At step 2 (Delete)	Total deleted products of the day will exceed 20 after the process	AIMS software prevents the action and require to reduce the product to be deleted	At step 1 (Delete)
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Table 1: Alternative flows of events for UC CUD product

6. Input data

No	Data fields	Description	Mandatory	Valid condition
1	Barcode		Yes	must be unique, not including special characters
2	Title		Yes	
3	Category	Choose from a list	Yes	
4	Original value		Yes	must be positive
5	Current value		Yes	must be positive
6	Description		No	
7	Height		Yes	must be positive
8	Width		Yes	must be positive
9	Length		Yes	must be positive
10	Weight		Yes	must be positive

Table 2: General data for UC Create product

No	Data fields	Description	Mandatory	Valid condition
1	Author		Yes	
2	Cover type	Choose from a list	Yes	
3	Publisher		Yes	
4	Publish date		Yes	
5	Number of pages		No	integer number, must be positive
6	Language		No	
7	Genre		No	

Table 3: Data for book type products creation

No	Data fields	Description	Mandatory	Valid condition
1	Chief editor		Yes	
2	Publisher		Yes	
3	Publish date		Yes	
4	Issue number		No	

5	Publication frequency		No	
6	Language		No	
7	ISSN	an 8-digit number	No	must be unique, contains 8 digits
8	Language		No	
9	Section		No	

Table 4: Data for newspapers type products creation

No	Data fields	Description	Mandatory	Valid condition
1	Artist		Yes	
2	Record labels		Yes	
3	Tracks list		Yes	
4	Genre		Yes	
5	Release date		No	

Table 5: Data for CD type products creation

No	Data fields	Description	Mandatory	Valid condition
1	Disc type	Choose from a list	Yes	
2	Director		Yes	
3	Runtime		Yes	integer number, must be positive
4	Studio		Yes	
5	Language		Yes	
6	Subtitles		Yes	
7	Release date		No	
8	Genre		No	

Table 6: Data for DVD type products creation

No	Data fields	Description	Mandatory	Valid condition
1	Barcode/ID		Yes	unique
2	Data field to be modified (similar to data meant for		Yes	depend on the data to be modified

	creating product)			
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Table 7: Data for UC Update product

No	Data fields	Description	Mandatory	Valid condition
1	List of product barcode/ID that will be deleted		Yes	

Table 8: Data for UC Delete product

7. Output data

No	Data fields	Description	Display format	Example
1	Barcode	ID of the product		202312345
2	Title	Title of a media product		DVD Phim Vượt ngục
3	Category	Type of product		DVD
4	Original value	Original price of the corresponding media product	Comma for thousands separator Positive integer Right alignment	123,000
5	Current value	Current price of the corresponding media product	Comma for thousands separator Positive integer Right alignment	130,000
6	Description	Describe what the product is about	Right alignment	
7	Height		Positive integer Right alignment	3 cm
8	Width		Positive integer Right alignment	10 cm
9	Length		Positive integer Right alignment	13 cm
10	Weight		Positive integer Right alignment	20 gr

Table 9: Output of general data for UC Update product

No	Data fields	Description	Display format	Example
1	Author			Nguyễn Nhật Ánh
2	Cover type			paperback
3	Publisher			Nhà xuất bản Trẻ
4	Publish date		dd/mm/yyyy	20/10/2015

Table 10: Output of data for book type products update

No	Data fields	Description	Display format	Example
1	Chief editor			Đỗ Huy Hoàng
2	Publisher			Báo Tiền Phong
3	Publish date		dd/mm/yyyy	15/3/2026

Table 11: Output of data for newspapers type products update

No	Data fields	Description	Display format	Example
1	Artist			Soobin
2	Record labels			SS Label
3	Tracks list			
4	Genre			Pop

Table 12: Output of data for CD type products update

No	Data fields	Description	Display format	Example
1	Disc type			Blu-ray
2	Director			
3	Runtime		Positive integer Right alignment	73 mins
4	Studio			
5	Language			
6	Subtitles			

Table 13: Output of data for DVD type products update

8. Postconditions

- Successful Create/Update: The product database is updated, and changes are reflected in the system.
- Successful Delete: Products are either removed or marked as "Deactivated," and the action is logged.